

TABUK, Saudi Arabia

WMO: 403750

Lat: 28.365N

Lon: 36.619E

Elev: 778

StdP: 92.33

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.1	3.8	-12.2	1.4	15.7	-10.1	1.7	16.1	11.7	18.4	10.2	18.4	1.1	110	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.5	41.8	18.9	40.2	18.4	39.1	18.1	20.9	36.5	20.0	35.6	19.3	35.4	4.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	11.9	27.1	14.1	11.1	27.1	13.1	10.3	26.6	64.2	36.4	60.6	35.6	58.1	36.5	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.5	7.1	DB	-0.3	43.4	1.5	1.1	-1.4	44.1	-2.3	44.8	-3.1	45.4	-4.2	46.2	(4)
(5)			WB	-2.9	21.6	1.4	2.1	-4.0	23.1	-4.8	24.3	-5.6	25.5	-6.7	27.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.7	11.7	14.0	17.8	22.6	27.1	30.2	31.9	32.1	29.4	24.7	17.9	12.9
	DBStd	7.84	3.06	3.39	3.66	3.51	2.88	1.95	1.84	1.89	2.09	2.91	3.18	2.93
	HDD10.0	26	16	4	0	0	0	0	0	0	0	0	0	6
	HDD18.3	615	208	127	56	6	0	0	0	0	0	1	46	172
	CDD10.0	4675	67	116	242	379	530	607	678	686	583	457	237	94
	CDD18.3	2223	1	6	39	136	272	357	419	427	333	199	32	2
	CDH23.3	31285	25	104	523	1639	3567	5236	6493	6575	4539	2216	333	34
	CDH26.7	18729	2	21	178	756	2027	3294	4223	4304	2747	1091	84	4
(14) Wind	WSAvg	2.8	2.3	2.8	3.1	3.3	3.4	3.3	3.3	3.2	2.8	2.5	2.1	2.0
(15) Precipitation	PrecAvg	30	10	1	5	1	3	0	0	0	0	3	3	3
	PrecMax	96	75	13	26	12	51	1	1	3	2	21	41	18
	PrecMin	3	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	24	17	3	8	3	10	0	0	1	0	6	8	4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	29.4	33.9	36.9	40.0	42.0	43.1	43.1	41.7	37.4	31.2	26.9
		MCWB	12.9	13.7	14.7	16.1	17.3	17.9	18.9	19.6	18.6	17.1	14.9	13.3
	2%	DB	23.2	26.4	31.0	34.9	38.1	40.1	41.2	41.9	39.2	35.8	29.0	23.9
		MCWB	11.6	12.5	13.7	15.8	16.8	17.7	18.8	19.4	18.1	16.7	14.3	12.2
	5%	DB	20.9	24.0	28.2	33.0	36.8	38.9	40.0	40.2	38.0	34.1	27.1	21.8
		MCWB	10.7	11.4	13.0	14.7	16.4	17.4	18.2	19.0	17.8	16.1	13.8	11.5
	10%	DB	18.6	21.8	26.1	31.1	35.1	37.8	38.8	39.0	36.8	32.2	25.0	19.7
		MCWB	9.7	10.5	12.1	14.1	15.8	17.2	17.9	18.7	17.5	15.8	13.2	10.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	15.4	16.5	18.5	20.2	21.2	21.8	22.4	21.3	19.4	17.1	15.6
		MCDB	21.8	24.1	28.8	33.2	34.5	37.4	36.5	38.2	35.4	31.8	25.8	22.4
	2%	WB	13.2	13.8	15.0	16.8	18.3	19.6	20.4	21.0	19.7	18.0	15.9	13.6
		MCDB	20.3	23.8	28.0	30.8	32.8	35.3	36.0	37.2	34.5	30.8	24.8	21.8
	5%	WB	11.4	12.4	13.7	15.7	17.1	18.4	19.5	20.1	18.7	17.2	14.9	12.1
		MCDB	19.5	22.3	26.3	30.5	33.7	35.1	36.1	36.0	34.0	30.3	23.6	19.7
	10%	WB	10.1	11.1	12.7	14.7	16.3	17.6	18.7	19.4	18.2	16.5	14.0	10.9
		MCDB	17.8	20.6	24.9	29.7	33.1	35.0	36.0	35.8	33.5	29.6	23.3	18.7
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	13.9	14.5	15.0	14.9	14.9	14.1	14.5	14.9	14.6	13.5	13.5
		MCDBR	16.1	16.8	16.8	16.6	16.1	15.9	15.0	15.3	15.8	16.1	15.7	15.5
	5% WB	MCWBR	8.4	8.1	7.2	6.7	5.9	5.3	4.6	4.7	5.1	6.0	6.8	7.9
		MCDWR	14.2	14.8	15.4	15.3	15.1	15.2	14.3	14.7	14.8	14.3	13.0	13.7
(40) Clear-Sky Solar Irradiance	taub		0.307	0.338	0.390	0.462	0.456	0.397	0.408	0.422	0.395	0.398	0.347	0.313
		taud	2.401	2.292	2.109	1.914	1.960	2.165	2.164	2.123	2.178	2.152	2.297	2.394
	Ebn at Noon		936	930	896	840	843	889	878	864	879	851	878	911
		Edh at Noon	100	121	155	195	187	152	151	156	143	138	110	96
(44) All-Sky Solar Radiation	RadAvg		4.18	5.15	6.37	7.24	7.83	8.40	8.08	7.52	6.71	5.48	4.40	3.88
	RadStd		0.19	0.22	0.17	0.17	0.30	0.09	0.17	0.16	0.18	0.18	0.17	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.62	N/A	-1.94	+0.80	N/A	N/A	N/A	-74	+241
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page